

A Case Study Exploring the Applicability of Heuristic Evaluation in Smart Home Systems

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ABSTRACT

Smart home systems, integrating multiple devices and interaction methods, give rise to complex usability challenges. While heuristic evaluation has been a preferred method for evaluating usability across various domains, its applicability to smart home systems remains underexplored. This case study investigates the strengths and constraints of heuristic evaluation applied to smart home systems. Our results suggest that while heuristic evaluation effectively identifies GUI issues and guides expert solutions, it falls short in addressing technical and hardware intricacies unique to hybrid systems like smart homes. This study not only presents recommendations for enhancing heuristic evaluations in this evolving domains, but also offers practical insights specifically for the evaluation of hybrid systems.

CCS CONCEPTS

• Human-centered computing → Heuristic evaluations.

KEYWORDS

Heuristic evaluation, smart home, user interface, usability, hybrid system

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1 INTRODUCTION

The adoption of smart home systems is continuously growing, with an expected increase of over 672 million households by 2027 [8]. At the current stage, smart home systems can conveniently control home electronics and appliances within the home space or remotely

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Table 1: Nielsen's 10 Usability Heuristics

No.	Description
HE1	Visibility of system status
HE2	Match between system and the real world
HE3	User control and freedom
HE4	Consistency and standards
HE5	Error prevention
HE6	Recognition rather than recall
HE7	Flexibility and efficiency of use
HE8	Aesthetic and minimalist design
HE9	Help users recognize, diagnose, and recover from error
HE10	Help and documentation

through mobile phones, tablets, or voice speakers and provide users with information about home facilities. Combining multiple devices into one bundled system, a smart home system should have worked as a cohesive unit. Nevertheless, additional usability problems arise, including cumbersome device initialization setups [27] and difficulties setting up automation rules or adding cross-device functionalities [6]. Consequently, it is crucial to identify usability problems when designing and developing smart home systems.

Introduced by Nielsen and Molich in the 1990s, heuristic evaluation is a method predicated on a set of guidelines for assessing usability problems (see table 1) [23, 25]. This method typically involves 3-5 evaluators who, utilizing established usability heuristics and drawing upon their individual expertise, identify potential issues. As a general evaluation method, due to its simplicity, efficiency, and high cost-effectiveness, heuristic evaluation has been widely used to evaluate various domains (e.g., websites [7, 26, 31], applications [2, 30], games [5, 15, 32], and buildings [1, 10]). Driven by diverse interaction technologies and trends towards more intuitive, user-centric interfaces, heuristic methods have expanded from evaluating web and mobile graphical user interface (GUI) to a broader range of interaction systems, including physical devices [35], voice interactions [14] and multimodal interactions [34, 38].

However, it is crucial to acknowledge that smart home systems are often hybrid in nature, integrating multiple functionalities to serve various users and connecting to a diverse array of devices. While most research related to heuristic evaluation in smart homes

has been concentrated on home control applications or smart devices, there is a gap in understanding their broader interconnectivity. The predominant focus has been on GUI usability issues [19, 20, 28], overlooking the more complex interplay of hardware and software. Nevertheless, smart home systems often enable interactions through GUI, voice user interface(VUI), or direct device manipulation [16]. Previous study [18] has explored the integration of mobile applications with voice speakers, but it did not extend to the wider context of device interconnectivity. The unique modality of each device, combined with multi-device linkages, culminates in a complex mixture of virtual and physical feedback. This intricacy poses significant challenges for heuristic evaluations, which are typically more apt for stand-alone products than for such complex hybrid systems.

In this case study, we aim to investigate the applicability of heuristic evaluation for assessing smart home systems. Also, we want to discuss what must be considered when performing heuristic evaluations of such hybrid systems. Taking the Xiaomi smart home system as an example, we conducted a heuristic evaluation study with usability experts. The heuristic evaluation of the smart home system followed three standardized processes. First, a brief session was conducted to ensure each evaluator fully understood the 10 heuristics. After that, the evaluators completed the evaluation tasks individually in the smart home experimental platform we provided and documented the usability issues according to the heuristics. Finally, through a debriefing session, participants proposed potential solutions to these problems and discussed the effectiveness of the heuristic evaluation. Our findings highlight that heuristic evaluation fully explores GUI issues and assists experts in proposing detailed solutions. However, it's hard for technical and hardware hybrid problems to correspond with the existing heuristic, which weakens the evaluation effect to some extent. We suggest that heuristic evaluation tasks should be tailored to the characteristics of the target user's usage phase and should be carried out in the actual living space, with the participation of usability experts from multiple backgrounds and with the experts' understanding of the user's experience in order to obtain better evaluation results.

The contribution of this case study is threefold: (1) Our study presents the effectiveness of heuristic evaluation in smart home systems, emphasizing its role in addressing GUI issues; (2) We find that the constraints of employing heuristic evaluation for smart home systems revolves around technical and hardware intricacies, primarily due to instances where technical problems are erroneously ascribed to hardware issues; (3) We share experience and lessons learned during the research process, providing recommendations for a better performance in this method to face a broader range of such hybrid systems.

2 METHOD

This case study aimed to explore the strengths and constraints of heuristic evaluations in assessing smart home systems. We used the set of 10 heuristics proposed by Nielsen, allowing evaluators to identify usability issues and propose solutions based on the heuristic.

Specifically, we recruited 5 evaluators, each with specialized knowledge in both smart home research and usability evaluation,

ensuring an optimal outcome using this method [21]. According to the concept of expert classification of Federico [3], evaluators were asked to self-rate their professional level based on hours of deliberate practice and university degree. The participating evaluators, on average, rated their expertise at 4.8 on a 5-point Likert scale (5 being the highest, 1="novice" and 5="expert"), and their tenure in smart home research averaged 2.3 years.

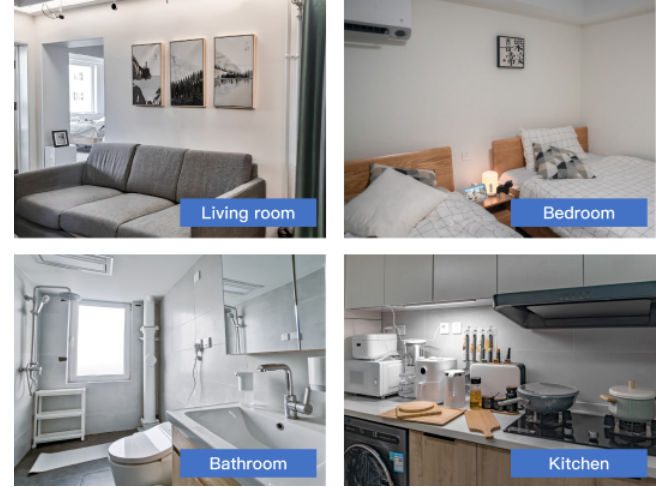


Figure 1: Integrated Experimental Platform for Smart Home

2.1 System Evaluation and Task Design

In alignment with our research scope, we focused on a consumer-grade system that facilitates seamless integration between the central management and smart hardware endpoints. Our Smart Home Research Group established the "Integrated Experimental Platform for Smart Home" in an apartment [11], which includes a living room, bedroom, study, kitchen, bathroom, and balcony, spanning roughly 60m² (see Figure 1). This space is designed for actual residential use, simulating real-life situations. All embedded appliances and devices within this environment are exclusively from the Xiaomi brand, amounting to 56 in total.

In this expansive and intricate smart home system, we narrowed the scope by specifying smart devices and setting evaluation tasks, making the assessment effective and detailed. (see Table 2)

We evaluated 7 commercially available smart home devices, considering functionality, capability (e.g. communication protocols, proprietary technologies, closed or open systems) and connectivity (internet, network and other requirements). These devices are lighting, camera, robot vacuum, voice assistant, smart switch, air purifier, and air tester. These devices encompass a spectrum of interfaces—GUI, VUI and physical interface—demonstrating the multi-modality inherent to smart home. Users have the capability to operate these devices via the Mijia app, serving as the central command hub.

We designed 9 evaluation tasks, covering both setup and automation aspects. The tasks related to setup investigate usability issues during individual device connection processes. Each device follows the 6 steps when initialized via the Mijia app: add, search,

Table 2: Nine Evaluation Tasks with Task Description and Smart Devices

No.	Smart Devices	Task	Task Description
1	Lighting	Setup	Follow these steps to setup device: 1.add device 2.search device 3.reset device 4.connect device 5.configuration 6.test device
2	Camera		
3	Robot Vacuum		
4	Voice Assistant		
5	Smart Switch		
6	Lighting	Automation	When leaving home, the lights automatically turn off.
7	Voice Assistant		When returning home, the Voice Assistant says, "Welcome home."
8	Voice Assistant		When saying "I'm feeling down," the Voice Assistant plays a song.
9	Air Purifier, Air Tester		When the indoor PM2.5 level exceeds 50 and the user enters Jinshu Road, the air purifier turns on.

reset, connect, configuration and test. Tasks related to automation explore usability issues in setting automation rules for interconnected multi-device systems. Users configure The Internet of Things (IoT) automation based on the Trigger-Action Programming (TAP) paradigm [37], in the form of "IF triggered, THEN operate."

2.2 Heuristic Evaluation Process

2.2.1 Brief Session. Teams embarking on heuristic evaluations require preliminary training and preparation to ensure they know exactly what they should do and cover during the evaluation [24].

A day prior to conducting the evaluation, we assembled 5 evaluators for a briefing session. Drawing from Jakob's 10 Usability Heuristics Posters [23], 5 evaluators revisited the heuristic evaluation, despite their previous experiences with the method. To ensure a thorough grasp during the evaluation, we asked each person to peruse and comprehend the heuristics, emphasizing familiarity with the 10 guidelines.

2.2.2 Evaluation Session. The evaluation session is conducted by having each evaluator individually inspect the interface [9, 13]. Prior to the commencement, each evaluator was handed out 2 sets of printed materials, with their usage duly elucidated (see figure 2).

space for evaluators to note down observed problems. Additionally, it elucidates a standardized scoring mechanism for severity, ensuring consistent adherence across the board.

- **Heuristic Evaluation Checklist:** It consists of reminders for each of the 10 heuristics, allowing evaluators to quickly find the corresponding one.

Initially, evaluators are asked to complete tasks one by one according to the description on the workbook, noting any usability issues encountered along the way.

Subsequently, evaluators go through the tasks again, but this time they apply relevant heuristics to previously identified issues. Evaluators can refer to the checklist to quickly find the corresponding heuristic.

Finally, when the evaluators have completed all tasks and listed the full set of discovered usability issues, we ask them to rate the severity of each issue. This is rooted in the observation that evaluators, when engrossed in uncovering novel usability challenges, may struggle to provide accurate severity assessments [22].

2.2.3 Debriefing Session. Within this session, we collated a complete list of identified issues. Conventionally, evaluators would suggest potential solutions to these issues grounded in heuristics. However, our objective spanned beyond just rectifying these issues; we aimed to discern both the strengths and constraints of heuristic evaluations within the context of smart homes. To this end, we recruited 2 actual users with experience living in our Smart Home Platform for more than 6 months. They reported the daily use of smart devices, stating that they use smart homes every day, using an average of 10.5 devices. They also possess dual professional knowledge in smart homes and usability evaluation. They rated their expertise at 4 on a 5-point Likert scale, and their tenure in smart home research averaged 1.2 years.

We first reviewed the previous two sessions, not only facilitate recollection for the 5 evaluators but also provide the 2 real users a glimpse into our procedures.

Next, we detailed each usability problem from the list of issues, weighing both its severity and frequency. We also verified with users whether the results of the heuristic evaluation were consistent with their actual experience.

Thirdly, each of the 7 participants proposed potential solutions for every problem, grounded in the heuristics. These solutions were subsequently grouped using affinity diagrams [29, 36].

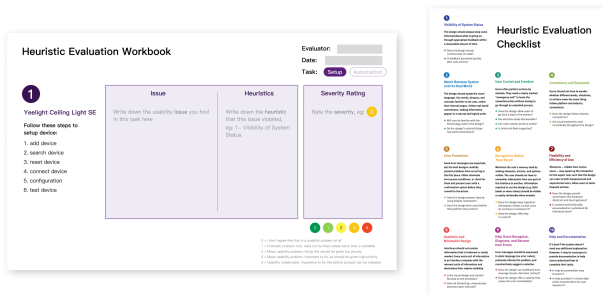


Figure 2: Printed Materials for Evaluators in Evaluation Session

- **Heuristic Evaluation Workbook:** We provided evaluators with a heuristic evaluation workbook designed for documenting evaluations. This includes task descriptions, identified issues, severity assessments, and heuristics, providing ample

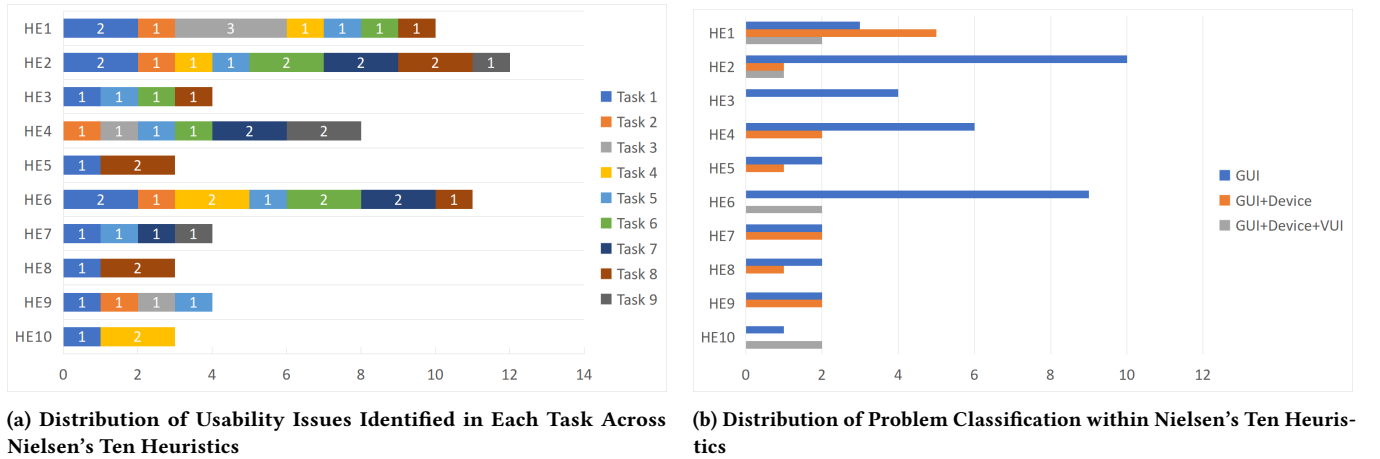


Figure 3: Distribution of Usability Issues Identified by Five Evaluators in Nielsen's Ten Heuristics in the Evaluation Session

In conclusion, we engaged in a comprehensive discussion with all participants, delving into the merits and drawbacks of utilizing heuristic evaluations in smart home contexts. We also discussed their actual feeling in practice and sought their insights on our research question.

2.3 Data Collection and Analysis

For the heuristic evaluation data, we initially classified the identified usability issues according to Nielsen's 10 heuristic principles. Subsequently, each issue was categorized into one of three groups: GUI, GUI+device, or GUI+device+VUI. We then quantitatively analyzed the data, contrasting the usability issues with the 10 heuristics, tasks, and problem classifications to examine the distribution of usability across the smart home system.

For the data from the discussion, we transcribed the text verbatim. Following the thematic analysis method [4], two researchers conducted open coding on the content. After all the coding was completed, the research team met and discussed any discrepancies in the codes until a consensus was reached. Through this, we determined the extent to which heuristic evaluation results mirrored users' actual usability problems and gauged evaluators' attitude towards heuristic evaluation.

Finally, we compared the quantitative data with the qualitative results to reach our research findings.

3 FINDINGS

3.1 Effective Evaluation of Usability Issues in Smart Home Systems

The heuristic evaluation, facilitated by 10 guiding principles and a structured process in three sessions, adeptly unpacks the intricacies of smart home systems, providing a certain extent of effective assessment of usability issues in smart home systems.

Figure 3 shows the distribution results of each task and problem classification within the heuristics. In the smart home systems we evaluated, the most frequently violated heuristics were HE2 *Match between system and the real world*, HE1 *Visibility of system status*,

and HE6 *Recognition rather than recall*. Intriguingly, these heuristics closely align with the systemic attributes of smart homes, echoing the inherent multi-device and multi-modal traits of modern smart homes that inevitably give rise to various usability problems.

From the points highlighted above, GUI problems predominantly align with HE2 and HE6. This sheds light on the fact that the current Mijia App, the central interface for all smart devices, employs terminology and metaphors that are not intuitively graspable, imposing cognitive burdens on users. Many problems arise from semantic naming in the GUI text, such as the button for setting automation rules as "create smart" rather than using more user-friendly and familiar terms.

When usability issues result from the combined actions of GUI and devices, they are primarily associated with HE1. This is due to the Setup phase involving tasks that require connecting multiple devices, and typically, multiple devices have multi-modal user interfaces. For instance, when connecting the voice assistant and the Mijia App in task 6, users need to simultaneously verify the system status of both devices. The sequential feedback from voice assistant's VUI and the Mijia App's GUI, attributed to communication nuances, tends to bewilder users about the authentic system state.

Additionally, for the same issue, even though evaluators have varied ways of describing the problem, they ultimately categorize it under the same heuristics. Taking task 1 of resetting the lighting switch 7 times as an example: some evaluators pinpoint it as semantic ambiguity in GUI issues, others attribute it to hardware design flaws, while another group perceives it's a problem in the interaction flow. Therefore, another advantage of heuristic evaluation is the thorough and comprehensive description of the same usability concerns in smart homes.

3.2 Comprehensive in Exploring GUI Issues

Heuristic evaluation performs best in evaluating GUI issues in smart homes. As shown in figure 4, the largest portion of issues identified by evaluators falls under GUI concerns. During this heuristic evaluation, we identified a total of 37 usability issues, with 27 being

specific to the GUI. The strengths of heuristic evaluation in the context of smart home GUI are as follows:

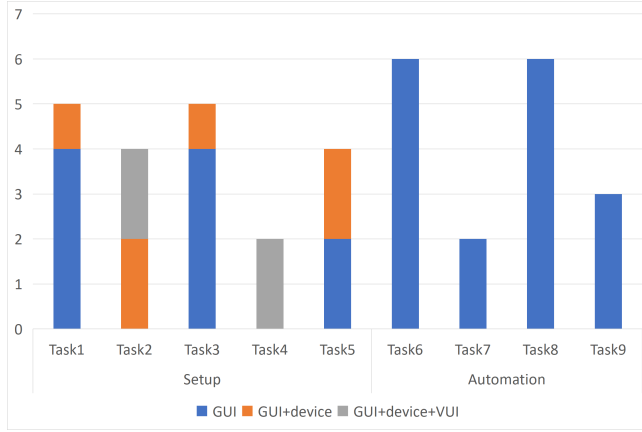


Figure 4: The Distribution of Problem Classification within Each Task

- **GUI issues are described comprehensively in detail, touching on aspects such as color, text, functions, and layout.** For instance, experts pointed out, "The gray circular element to the left of the *if* and *then execute* text is easily mistaken as a selection button." Evaluators agree that the more focused scope of such issues makes it easier for them to focus on details. In contrast, as the issues integrate physical devices and voice interactions, the descriptions became notably broader. For instance, when addressing GUI+device+VUI issues, they stated, "When resetting the voice assistant, the VUI indicates a successful connection, but Mijia app shows a connection failure." Due to the complexity of smart homes, evaluators can't discern where the exact issue lies, hence they document these problems using more generalized descriptions.
- **Through GUI issues, the inherent limitations of the current TAP paradigm can be identified.** By looking at the distribution of problem classification within each task, we find issues related to automation tasks fall under GUI category. The present automation configuration mechanism and processes are somewhat isolated from the actual interactions of smart homes, failing to fully utilize contextual information and human behaviors within the IoT ecosystem to promote intuitive and cost-effective programming [17].
- **The solutions for GUI issues possess greater consistency.** Evaluators' suggestions related to GUI facilitate categorization in the affinity diagram, allowing for universally agreed-upon solutions rapidly. However, the solutions proposed for GUI+device+VUI issues reveal significant variations among evaluators.

3.3 Technical Issues Cannot be Mapped to Existing Heuristics

A weakness in applying heuristic evaluation to smart home systems is that, in certain situations, technical issues cannot be mapped to

the corresponding heuristics. A consensus among evaluators is that heuristic evaluation tends to spotlight design problems predominantly.

During evaluation, evaluators encountered issues that couldn't be categorized under Nielsen's 10 heuristics. For instance, they noted that during the setup, using the "auto-search" failed to detect certain device, necessitating "manual search". They were uncertain whether it was an issue with the smart device or a network connection. Echoing this sentiment, real users also reported facing such issues. Due to the unclear root causes, these types of problems are often the most challenging to resolve in real-life scenarios. At times, network connectivity might be the culprit behind usability issues, yet users tend to reboot the device hoping for the system to operate normally again.

In essence, when confronted with technical issues, the inherent constraints of heuristic evaluation can potentially dilute its evaluative efficacy, leading to scenarios where network disruptions are mistakenly ascribed to device usability issues.

4 DISCUSSION

4.1 Enhancing the Evaluation with Multidisciplinary Experts

Based on our findings that the heuristics can evaluate GUI problems well but do not achieve good results for VUIs and the device itself, and there are still some technical problems that cannot be categorized within the 10 heuristics. Through discussions with experts, GUI problems are more intuitive, and easier to propose detailed solutions. On the other hand, when facing usability problems arising from the combination of hardware and software, since technical reasons cause usability problems, it is difficult for experts to quickly attribute the problems and give solutions due to the need for more understanding of the technical problems. Even if a solution is generated, the feasibility of the solution needs to be clarified. For example, to propose a solution for resetting the device by re-switching the lights on and off 7 times in Task 1, some evaluators have proposed using a long press to reset the device, but it is not sure whether the hardware supports such an operation.

At this stage, the smart home system defined in this paper is a bundled system, i.e., smart devices can be bundled together and programmable [33]. In the future, smart home systems will be able to learn user habits and adjust and adapt to user needs based on user behavior, ultimately reaching the level of predictable user behavior [33]. When Artificial Intelligence (AI) is integrated into a smart home system, the system's technical complexity will increase, leading to expert evaluation becoming more complex than the current stage. Therefore, selecting usability experts with design or technical backgrounds to discuss together in the problem evaluation and solution development phases will help identify and solve usability problems of smart home systems more effectively.

4.2 Empathizing with Diversified Users

The heuristic evaluation examines usability problems by experts based on their knowledge and experience. However, the heuristic evaluation process does not capture the variability of different user groups in real-home environments. Elderly, disabled, children, and users with different technical proficiency may encounter usability

problems that vary greatly [12]. During the evaluation process, experts can only identify a limited number of usability problems based on their physical and cognitive characteristics. Furthermore, when it came to proposing resolutions, the lack of targeted solutions meant that experts often suggested a range of solutions, but were unable to determine the optimal one.

As an illustrative instance, when an evaluator proposes a solution to the analyzed problem, choosing appropriate inputs and outputs requires thoughtful consideration tailored to specific user groups. The evaluator stressed that older individuals may prefer VUI, whereas younger counterparts may suffice with subtle physical light prompts or GUI reminders. This underscores the imperative of catering to diverse user preferences and demographic characteristics in designing and implementing solutions within usability assessments.

4.3 Conducting On-site Research was Beneficial

The usability evaluation of our smart home system unfolded in a natural home scenario, which helped us to check out the usability issues associated with the space. For the Automation tasks, we asked the experts to connect the camera on the living room's ceiling to the Mijia APP. The camera, which was already mounted on the ceiling, could not be removed due to the need to scan the QR code on the body's base, leading us to remove a camera from the kitchen tabletop for the expert to complete the task. Similar space-related usability issues are more accessible in natural home scenarios.

In addition, regarding the defects arising from the use of automation rules, it is also tricky for experts to judge the usability issues by individual tasks when evaluating them, e.g., the system is unable to determine whether the user is temporarily out of the house or working away from home, and therefore starts cleaning as soon as it judges that the user is out of the house, which is not in line with the user's understanding and expectation of away-from-home modes. The dynamic scenarios of smart homes present challenges for heuristic evaluation, and it is difficult to construct comprehensive dimensions to evaluate usability problems on a task-by-task basis. Therefore, heuristic evaluation in natural home scenarios should allow experts to experience or restore the user's usage scenarios through storyboarding. This facilitates experts to understand the tasks better and fully uncover usability issues.

4.4 Exploring Usability Challenges in Hybrid Systems

Hybrid systems converge various technologies, accommodating multiple users, devices, and modalities. As smart home systems target diverse user groups, application scenarios evolve with user demands and iterative device updates. The above system characteristics necessitate comprehensive considerations when conducting heuristic evaluations. Incorporating IoT and AI in smart transportation may introduce usability challenges, demanding expertise from diverse backgrounds to validate solutions. Similarly, addressing usability in education platforms, catering to users with varying ages and comprehension abilities, requires empathizing with users. Retail inventory management systems, influenced by differing physical spaces and device layouts in stores, benefit from integrating usability research with spatial considerations.

Moreover, in light of our research results, we discovered that when formulating heuristic evaluation tasks for hybrid systems, taking smart home systems as an example, focusing solely on instrumental and functional aspects of system usability is far from sufficient. Various usability issues may arise during actual usage, involving a broader spectrum of User Experience (UX) categories. Previous research indicates that users encounter usability issues at different stages, namely (1) installation, (2) usage, (3) routinization, (4) modification, and extension [39], when using smart home products. Discussions with users further corroborate this viewpoint, as users point out that heuristic evaluation tasks may need to catch up in covering typical usage scenarios, especially during prolonged usage, due to changes in user needs across different usage stages.

Therefore, in heuristic evaluations of hybrid systems, it is crucial to assess characteristics related to individual users and those linked to multi-users, multi-modalities, and multi-devices. This necessitates integrating insights from real user experiences and existing research literature on UX. Engaging usability experts with diverse backgrounds in field surveys is essential for comprehensively evaluating usability issues in hybrid systems. This collaborative approach enriches the assessment with nuanced perspectives.

Limitations. In this case study, we focused on usability issues of the Xiaomi's smart home system. It would be beneficial to examine problems of other brands of smart home control hubs, smart devices, and hybrid systems in other smart spaces, to understand the applicability of the insights presented and to discover more opportunities for heuristic evaluation. What else, our Smart Home Experimental Platform has only been lived by individuals, and we may have yet to explore the multiple residents usability problems. Therefore, future research could include the case of multiple users living under the same roof.

5 CONCLUSION

The solution to usability problems in smart home systems is the cornerstone of improving the system's user experience. This case study described the applicability and task design strategy of heuristic evaluation applied to smart home systems. This study showed that heuristic evaluation can uncover usability problems for GUI, VUI, and smart devices of smart home systems. It can provide optimal help for GUI problem discovery and solution development. The results of the investigation also showed that inviting usability experts from multiple backgrounds to participate in the usability evaluation based on the characteristics of the target users, the actual usage environment, and the usage phases of the users can more adequately uncover the usability problems, which provided suggestions and insights for researchers and practitioners to apply heuristic evaluation to evaluate smart home systems in the future.

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